INSTITUTIONAL

PORTFOLIO RISK DASHBOARD

Module architecture, feature definitions and subfeature explanations

15 MODULES | 130 TOP-LEVEL FEATURES | 520 EXPLAINED SUBFEATURES

PROJECT SCOPE

Asset class	Equities
Research frequency	Order-book level
Execution scope	Live-capable integration
Deployment	Local
Engineering model	Solo developer
Architecture	Modular services with shared platform infrastructure

Document purpose

This document explains the proposed functional scope of the Institutional Portfolio Risk Dashboard. It defines the responsibilities of each module, the behavior of each top-level feature and the meaning of its subfeatures. It is a functional planning specification, not a claim that the modules have already been implemented.

Version 1.0 | 31 October 2026

Document guide

The specification contains 15 modules and 130 top-level features, organized into five functional layers. Each module section defines its purpose, inputs, outputs and detailed feature/subfeature behavior.

Layer	Modules	Scope
Foundation & data operations	M01-M04	Feed configuration, ingestion, validation and reference data
Portfolio intelligence	M05-M06	Positions, valuation, exposure and allocation
Risk intelligence	M07-M10	Core risk, factors, microstructure liquidity and stress testing
Risk controls	M11-M12	Limits, pre-trade checks, alerts and escalation
User interface & governance	M13-M15	Dashboards, reporting, audit, APIs, security and infrastructure

Scope and design principles

Deterministic calculations: The same portfolio snapshot, market data and model configuration should reproduce the same result, subject to explicitly controlled stochastic seeds.

Point-in-time lineage: Every result should identify its portfolio timestamp, market-data sources, quality state, model version and calculation parameters.

Data quality before analytics: Stale, incomplete or invalid observations must be flagged and handled through declared policies; they must not silently enter risk calculations.

Separation of responsibilities: The backend owns calculations and controls. The dashboard visualizes results and submits authorized requests; it is not the source of truth.

Live-capable, not automatically live: Execution integration is designed for future live use, but production order routing requires separate broker/exchange connectivity, permissions, testing and operational approval.

Explainable control decisions: Limit checks and pre-trade decisions should return explicit reason codes and the inputs used to reach the decision.

M01 | Feed Registry & Configuration

Provides the control plane for registering market-data feeds and managing how the application connects to, subscribes to, monitors and changes them. It separates feed definitions from ingestion code so providers and connection settings can be changed without rewriting downstream risk services.

Inputs: Provider and exchange metadata; endpoint and protocol details; credential references; market-session calendars; operator configuration.

Outputs: Validated feed definitions, runtime connection settings, subscription instructions, health state and configuration audit history.

8 top-level features

M01.01 Feed Registry

Creates a canonical record for each external or internal data feed. The registry identifies what a feed supplies, who provides it, how it is accessed and which instruments or venues it covers.

- **Feed registration:** Create, edit, deactivate and uniquely identify a feed record.
- **Provider metadata:** Store provider name, environment, support details, protocol and entitlement notes.
- **Supported asset classes:** Declare supported instruments, exchanges, venues and data types.
- **Feed status:** Expose lifecycle states such as configured, connected, degraded, disabled or failed.

M01.02 Feed Configuration

Stores the parameters required to establish and operate a feed connection. Sensitive values should be held as secret references rather than copied into ordinary configuration records.

- **Connection parameters:** Define host, port, protocol, timeout, heartbeat and connection mode.
- **Endpoint configuration:** Maintain provider endpoints for development, test and production-like environments.
- **Authentication references:** Point to secret-store entries or environment variables without persisting raw credentials.
- **Subscription settings:** Specify instruments, channels, depth levels, update modes and requested data fields.

M01.03 Feed Lifecycle Management

Controls the operational lifecycle of each feed and provides consistent state transitions for the ingestion layer.

- **Start and stop:** Request orderly connection startup and shutdown.
- **Restart and recovery:** Restart a failed or unhealthy consumer using bounded retry rules.
- **Enable and disable:** Allow operators to temporarily exclude a feed without deleting its configuration.
- **Health state transitions:** Track configured, starting, live, degraded, stopping and failed states with timestamps.

M01.04 Feed Scheduling

Coordinates periodic or session-aware work for feeds that are polled or activated only during defined market windows.

- **Polling intervals:** Configure polling cadence by provider, endpoint or data category.
- **Refresh frequency:** Set refresh intervals for reference data, snapshots and slower-changing sources.
- **Session schedules:** Respect exchange open, close, auction and holiday calendars.
- **Schedule overrides:** Support controlled one-off changes for maintenance or exceptional sessions.

M01.05 Feed Priority Management

Defines precedence among multiple sources and determines which source should be used when the primary source is delayed or unavailable.

- **Primary and secondary feeds:** Declare preferred and backup providers for a given data type.
- **Fallback configuration:** Specify when and how the system switches to a backup source.
- **Priority rules:** Resolve competing observations using source rank, freshness and quality status.
- **Failback policy:** Define when a recovered primary feed can safely become authoritative again.

M01.06 Feed Monitoring

Publishes operational health indicators so that stale or disconnected feeds can be detected before they affect portfolio calculations.

- **Connection status:** Expose current connection and consumer state.
- **Latency monitoring:** Measure message arrival delay and processing delay where timestamps permit.
- **Last successful update:** Record the last valid message or successful poll time.
- **Error tracking:** Classify connection, authentication, subscription, parsing and provider errors.

M01.07 Configuration Versioning

Preserves a history of configuration changes so feed behavior can be reproduced and safely rolled back.

- **Configuration history:** Store who changed a setting, when it changed and the reason.
- **Version comparison:** Show field-level differences between saved versions.
- **Rollback:** Restore a known-good configuration after validation.
- **Effective configuration snapshot:** Attach the applied configuration version to feed and risk-run metadata.

M01.08 Feed Access Control

Restricts who can view or change feed definitions and sensitive connection settings.

- **Role permissions:** Separate read, configure, operate and administer permissions.
- **Credential references:** Limit visibility of secret names and prevent credential disclosure.
- **Configuration access logs:** Record configuration reads and mutations where required for operational review.
- **Change approval hooks:** Allow sensitive changes to require review before activation.

M02 | Market Data Ingestion Engine

Receives external market observations and converts them into a consistent internal event stream. The module must tolerate bursty traffic, reconnects and partial provider outages without silently losing data.

Inputs: Feed configuration; instrument subscriptions; exchange/vendor messages; historical files or API responses.

Outputs: Normalized market events, ingestion acknowledgements, checkpoints, raw-event archives and ingestion health metrics.

8 top-level features

M02.01 Real-time Price Ingestion

Consumes current market quotes and price updates used for valuation, exposure and risk calculations.

- **Last traded price (LTP):** Capture the latest reported execution price with source and event time.
- **Bid and ask:** Capture best bid and ask prices and displayed quantities.
- **OHLCV:** Ingest open, high, low, close and volume bars where supplied by the source.
- **Quote snapshots:** Support point-in-time quote snapshots for valuation and reconciliation.

M02.02 Order Book Ingestion

Captures visible market depth so liquidity, spread, imbalance and market-impact risk can be estimated.

- **Level 1 depth:** Capture the best bid and ask levels.
- **Level 2 market depth:** Capture multiple price levels on each side of the book, subject to feed entitlement.
- **Incremental updates:** Apply add, change and delete events to maintain a current book.
- **Full snapshots:** Load a complete book image to initialize or resynchronize local state.

M02.03 Trade Data Ingestion

Captures executed transactions and associated trade attributes for volume, turnover and realized execution analysis.

- **Executed trades:** Capture trade price, quantity, venue and event timestamp.
- **Trade volume:** Aggregate traded quantity over configurable intervals.
- **Trade value:** Calculate turnover using price multiplied by quantity, subject to currency and contract conventions.
- **Trade conditions:** Preserve sale conditions or flags that affect whether a print is eligible for analytics.

M02.04 Historical Data Ingestion

Loads past market observations to support backtesting, risk estimation, calibration and historical scenario replay.

- **Historical prices:** Import adjusted and unadjusted prices with adjustment metadata.
- **Historical trades:** Load transaction-level or aggregated trade records when available.
- **Order-book snapshots:** Import archived depth snapshots or event sequences for microstructure research.
- **Incremental backfill:** Fill missing date ranges without duplicating already accepted records.

M02.05 Multi-source Ingestion

Connects more than one provider or exchange while keeping source identity and data entitlements explicit.

- **Exchange feeds:** Support exchange-native or exchange-distributed market data.
- **Broker APIs:** Connect broker market-data endpoints when permitted by the broker agreement.
- **Data vendors:** Integrate licensed vendor feeds through provider-specific adapters.
- **Source tagging:** Retain provider, venue, channel and sequence metadata on each observation.

M02.06 Streaming Infrastructure

Moves high-frequency messages through the system using asynchronous consumers and bounded buffering.

- **WebSocket consumers:** Maintain streaming sessions and decode incoming messages.

- **Asynchronous ingestion:** Separate network I/O from downstream persistence and processing.
- **Buffering:** Absorb short bursts while enforcing memory and queue limits.
- **Backpressure:** Slow or shed noncritical work safely when consumers fall behind; never hide data loss.

M02.07 Data Normalization

Maps provider-specific payloads into canonical internal schemas and consistent units.

- **Timestamp normalization:** Convert source times to a canonical timezone and precision while preserving original timestamps.
- **Instrument identifiers:** Map vendor symbols and venue codes to internal instrument IDs.
- **Standardized schemas:** Represent quotes, trades, bars and book events with stable field names and types.
- **Unit normalization:** Standardize quantity, price scale, currency and lot-size conventions where possible.

M02.08 Ingestion Recovery

Restores ingestion after network interruptions, process restarts or detected sequence gaps.

- **Reconnect logic:** Re-establish sessions with exponential backoff and bounded retry policies.
- **Retries:** Retry transient failures while preventing unbounded retry storms.
- **Checkpointing:** Persist sequence positions or timestamps needed to resume safely.
- **Gap recovery:** Request missing ranges or snapshots and mark unrecoverable gaps explicitly.

M03 | Market Data Quality & Validation

Screens incoming data before it can influence valuation or risk. Quality decisions must be explicit, observable and reproducible; questionable observations should be quarantined or flagged rather than silently rewritten.

Inputs: Raw and normalized quotes, trades, bars and order-book events; instrument reference data; exchange sessions; source metadata.

Outputs: Validated observations, quality flags and scores, quarantined records, exception events and data-quality reports.

8 top-level features

M03.01 Missing Data Detection

Identifies absent observations and coverage gaps that may distort prices, returns or liquidity estimates.

- **Missing ticks:** Detect absent quote or trade updates relative to expected activity or sequence continuity.
- **Missing candles:** Identify absent time bars in a defined trading session.
- **Missing order-book updates:** Detect sequence gaps or stale books that may no longer represent current depth.
- **Coverage windows:** Report expected versus observed data coverage for each instrument and source.

M03.02 Price Validation

Flags prices that are structurally invalid or implausible relative to known market context.

- **Invalid prices:** Reject null, nonnumeric, nonpositive or out-of-range prices when the instrument convention requires otherwise.
- **Zero prices:** Apply instrument-aware rules to zero-valued observations rather than accepting them blindly.
- **Extreme price movements:** Flag jumps beyond configurable thresholds or exchange price bands.
- **Reference-price comparison:** Compare observations with prior close, official reference or independent source values.

M03.03 Timestamp Validation

Ensures event ordering and freshness can be trusted in downstream calculations.

- **Out-of-order events:** Detect messages arriving earlier than already processed sequence or event time.
- **Clock drift:** Compare source and receive clocks where available and flag material divergence.
- **Stale timestamps:** Identify observations too old for the intended real-time use.
- **Session alignment:** Check whether timestamps fall within valid exchange sessions or expected auction windows.

M03.04 Order Book Validation

Checks internal consistency of displayed depth before deriving spread, imbalance or liquidity metrics.

- **Crossed books:** Flag bid prices above ask prices, considering venue and feed semantics.
- **Locked books:** Identify equal best bid and ask prices and classify whether the condition is valid for the venue.
- **Invalid quantities:** Reject negative, nonnumeric or otherwise invalid displayed quantities.
- **Book continuity:** Check sequence continuity and require resynchronization after material gaps.

M03.05 Duplicate Detection

Prevents repeated messages from inflating turnover, volume or event counts.

- **Duplicate trades:** Use trade IDs, sequence numbers or composite keys to identify repeated prints.
- **Repeated market events:** Detect replayed quote or book updates where event identity is available.
- **Idempotent ingestion:** Ensure retrying an already accepted event does not create a second business record.
- **Duplicate audit:** Retain counts and reasons for duplicates removed or ignored.

M03.06 Data Consistency

Compares observations across sources and related fields to discover mismatches.

- **Cross-feed comparison:** Compare simultaneous quotes or prices from eligible sources.
- **Price discrepancy detection:** Measure absolute and percentage divergence against configurable tolerances.
- **Field consistency:** Check relationships such as high >= low and bid/ask quantity non-negativity.
- **Conflict resolution:** Apply documented source precedence or flag conflicts for manual review.

M03.07 Quality Scoring

Summarizes data fitness for use and allows downstream consumers to apply risk-aware acceptance rules.

- **Feed reliability scores:** Score availability, timeliness and error rates over a rolling window.
- **Completeness scores:** Measure the proportion of expected records or fields actually received.
- **Freshness scores:** Quantify age relative to the required update cadence.
- **Use eligibility:** Classify data as accepted, accepted-with-warning, restricted or rejected for defined calculations.

M03.08 Exception Management

Provides controlled handling for suspect records and enables investigation or correction.

- **Quarantine:** Move invalid observations out of the trusted data path while preserving original payloads.
- **Rejection:** Reject records that violate hard schema or market rules with a reason code.
- **Correction:** Apply controlled corrections with before/after values and provenance.
- **Replay:** Reprocess repaired or late-arriving records using idempotent logic.

M04 | Instrument & Reference Data Management

Maintains the canonical security and market-structure context required to interpret prices, positions and corporate events. Every instrument should have a stable internal identity even when vendor symbols differ.

Inputs: Exchange/security master files; vendor symbol mappings; corporate-action notices; trading calendars; currency and classification data.

Outputs: Canonical instrument records, identifier mappings, trading calendars, corporate-action events and reference-price metadata.

7 top-level features

M04.01 Instrument Master

Maintains one authoritative record for each tradable security or instrument covered by the platform.

- **Symbol:** Store exchange-facing ticker or trading symbol.
- **ISIN:** Store the International Securities Identification Number when applicable.
- **Exchange:** Identify the listing or trading venue.
- **Instrument ID:** Assign a stable internal key used throughout positions, market data and analytics.

M04.02 Security Classification

Organizes instruments into useful categories for concentration and factor analysis.

- **Sector:** Assign a broad economic sector classification.
- **Industry:** Assign a more granular industry classification.
- **Asset category:** Distinguish common equity, ETF, preferred security and other supported types.
- **Classification history:** Preserve effective dates when classifications change.

M04.03 Corporate Actions

Represents events that change security quantities, prices or shareholder entitlements.

- **Dividends:** Record cash or stock dividend terms and relevant dates.
- **Splits and bonuses:** Adjust share quantities and reference prices using event ratios.
- **Mergers:** Represent conversion, cancellation or successor-security relationships.
- **Adjustment lineage:** Keep original and adjusted values traceable to the applied event.

M04.04 Trading Calendar

Defines when each venue is open and which session rules apply.

- **Exchange holidays:** Maintain non-trading days and special closures.
- **Market sessions:** Define regular, pre-open, auction and post-market windows where relevant.
- **Trading hours:** Store local session times with timezone handling.
- **Calendar exceptions:** Support early closes or emergency schedule changes.

M04.05 Price Reference

Stores reference values used for validation, valuation and daily P&L calculations.

- **Previous close:** Maintain the prior official close for a given session.
- **Official closing price:** Store venue-published close and its source.
- **Reference price:** Store auction, settlement or other applicable reference values.
- **Currency and adjustment basis:** Record whether values are raw, adjusted or converted.

M04.06 Instrument Lifecycle

Tracks changes in the tradability and status of instruments.

- **Listing:** Record first trading date and listing venue.

- **Suspension:** Track temporary trading halts or suspensions.
- **Delisting:** Record last trading date and status transition.
- **Status-effective dates:** Ensure historical queries use the instrument state valid at that time.

M04.07 Instrument Mapping

Resolves differing identifiers used by exchanges, brokers and vendors.

- **Exchange symbols:** Map venue-specific tickers to the canonical instrument.
- **Vendor symbols:** Map provider-specific codes to the same canonical ID.
- **Internal identifiers:** Expose stable IDs to APIs and storage.
- **Mapping validation:** Detect ambiguous, stale or conflicting mappings before ingestion.

M05 | Portfolio & Position Management

Maintains the portfolio book: holdings, cash, transactions and valuation state. It is the source of truth for what the portfolio owns, while recognizing that broker or custodian records may need reconciliation.

Inputs: Portfolio definitions, trades and adjustments, broker/custodian files, instrument master, market prices and corporate actions.

Outputs: Current and historical positions, cash balances, valuation snapshots, realized/unrealized P&L and reconciliation exceptions.

8 top-level features

M05.01 Portfolio Creation

Creates and classifies the portfolios whose exposures and risks will be measured.

- **Portfolio registration:** Assign a unique ID, name, base currency and lifecycle status.
- **Portfolio classification:** Define strategy, mandate, benchmark and ownership metadata.
- **Portfolio hierarchy:** Support parent, sleeve or strategy-level groupings where required.
- **Portfolio settings:** Store valuation calendar, accounting conventions and applicable limits.

M05.02 Position Management

Tracks security quantities and direction at the portfolio and account level.

- **Long positions:** Represent positive holdings and their quantities.
- **Short positions:** Represent borrowed or sold-short quantities with explicit sign and borrow metadata where available.
- **Quantities:** Maintain settled, unsettled and available quantities when source data supports them.
- **Position events:** Apply trade, transfer, split and adjustment events in a deterministic order.

M05.03 Position Valuation

Calculates the value and profit/loss of each holding using an identified price source.

- **Market value:** Calculate quantity multiplied by an eligible mark price, with currency handling.
- **Cost basis:** Track acquisition cost using a declared accounting method.
- **Unrealized P&L:** Compare current valuation with cost basis.
- **Realized P&L:** Calculate profit or loss for quantities closed during a period.

M05.04 Cash Management

Tracks cash available to the portfolio and the effect of pending activity.

- **Available cash:** Calculate cash currently available for new activity under configured rules.
- **Reserved cash:** Track cash earmarked for unsettled trades, fees or other commitments.
- **Cash balances:** Maintain balances by currency and account.
- **Cash movements:** Record deposits, withdrawals, interest, dividends and fees.

M05.05 Transaction Management

Stores the events that change positions and cash, with enough detail to reproduce the book.

- **Buys and sells:** Record executions with quantity, price, fees, currency and execution time.
- **Transfers:** Represent inbound and outbound security or cash movements.
- **Adjustments:** Support controlled manual corrections with reason and approval metadata.
- **Trade lifecycle:** Track pending, partially filled, filled, cancelled and corrected transactions.

M05.06 Portfolio Reconciliation

Compares the internal book with broker, custodian or administrator records.

- **Broker statements:** Import external position and transaction statements.

- **Internal records:** Compare external records against the platform ledger.
- **Position mismatches:** Identify quantity, price, cash or instrument mapping differences.
- **Exception resolution:** Assign, explain and close differences while preserving evidence.

M05.07 Portfolio History

Preserves historical book states for time series, audit and reproducible risk runs.

- **Historical positions:** Query holdings as of a selected effective time.
- **Daily snapshots:** Persist end-of-day or configured intraday portfolio states.
- **Change history:** Track the events that explain position movement.
- **Point-in-time reconstruction:** Rebuild the portfolio as it was known at a prior timestamp.

M05.08 Multi-portfolio Management

Supports analysis across individual portfolios and consolidated groups.

- **Portfolio groups:** Create logical groups such as strategy, desk or mandate.
- **Consolidated holdings:** Aggregate holdings while retaining source-portfolio lineage.
- **Cross-portfolio netting views:** Show gross and net exposures across selected portfolios without losing individual views.
- **Shared reference settings:** Reuse approved benchmarks, currencies and risk conventions.

M06 | Exposure & Allocation Analytics

Converts positions and marks into measures of capital deployment and concentration. It explains where the portfolio is exposed, independent of whether those exposures are currently profitable.

Inputs: Valued positions, instrument classifications, portfolio hierarchy, cash balances and FX/reference data.

Outputs: Gross/net exposure, weights, sector and instrument breakdowns, concentration measures and limit-utilization data.

8 top-level features

M06.01 Gross Exposure

Measures the total absolute market exposure before offsetting long and short positions.

- **Long exposure:** Sum the market value of long positions.
- **Short exposure:** Sum the absolute market value of short positions.
- **Gross exposure:** Add long and short absolute exposures.
- **Gross exposure ratio:** Express gross exposure relative to NAV or another configured denominator.

M06.02 Net Exposure

Measures directional exposure after combining long and short positions.

- **Net market value:** Subtract short exposure from long exposure.
- **Net exposure ratio:** Scale net exposure by NAV or another defined base.
- **Directional interpretation:** Show whether the portfolio is net long, net short or approximately neutral.
- **Currency-aware netting:** Avoid invalid offsets across currencies unless converted under a documented FX convention.

M06.03 Sector Exposure

Aggregates portfolio exposure by sector and industry classification.

- **Sector aggregation:** Sum position market values and weights by sector.
- **Industry aggregation:** Provide a more granular breakdown within sectors.
- **Unclassified exposure:** Show positions with missing or unresolved classifications.
- **Relative exposure:** Compare portfolio sector weights with a chosen benchmark where available.

M06.04 Instrument Exposure

Shows position-level exposure and identifies dominant single-name risks.

- **Security market value:** Display marked value for each instrument.
- **Portfolio weight:** Calculate each holding as a share of NAV or gross exposure.
- **Ranked holdings:** Sort largest exposures and short positions.
- **Exposure change:** Compare current exposure with prior snapshots.

M06.05 Geographic Exposure

Aggregates risk by country or region using a configured issuer or revenue classification.

- **Country mapping:** Assign an issuer domicile or other defined geographic attribute.
- **Regional grouping:** Roll countries into regions for summary analysis.
- **Revenue geography:** Optionally distinguish listing domicile from revenue-source geography.
- **Unknown geography:** Keep unmapped exposure visible rather than dropping it.

M06.06 Concentration Analytics

Quantifies dependence on a small number of holdings or groups.

- **Top holdings:** Report the largest positions and their cumulative share.
- **Concentration ratios:** Calculate measures such as top-five or top-ten weight.

- **Herfindahl-Hirschman Index (HHI):** Sum squared portfolio weights under a stated weighting convention.
- **Concentration trend:** Track whether portfolio concentration is rising or falling.

M06.07 Portfolio Allocation

Presents how capital and portfolio weight are distributed across holdings and cash.

- **Capital allocation:** Show invested capital and cash allocation.
- **Percentage weights:** Calculate normalized weights using a declared NAV denominator.
- **Allocation breakdowns:** Slice allocations by security, sector, region or strategy.
- **Allocation drift:** Compare current weights with target or prior weights.

M06.08 Exposure Limits

Evaluates concentration and allocation against configured exposure thresholds.

- **Instrument limits:** Set maximum permitted exposure by security.
- **Sector limits:** Set sector or industry caps.
- **Portfolio-level thresholds:** Set gross, net or regional exposure limits.
- **Utilization reporting:** Show current exposure, limit, headroom and breach status.

M07 | Core Portfolio Risk Engine

Computes the principal statistical and market-risk measures for portfolios and individual positions. Every result should include its valuation timestamp, lookback window, confidence level, model method and input-data lineage.

Inputs: Historical and current market observations, portfolio snapshots, returns, benchmark series, risk-free rate and model configuration.

Outputs: Volatility, VaR, Expected Shortfall, drawdown, correlation/covariance, beta, risk-adjusted metrics and contribution measures.

10 top-level features

M07.01 Historical Volatility

Estimates the variability of returns over a selected lookback window.

- **Rolling volatility:** Recalculate standard deviation on moving windows.
- **Return frequency:** Support daily and intraday returns only where data quality and sampling assumptions are suitable.
- **Annualized volatility:** Scale period volatility using an explicitly configured trading-period convention.
- **Window comparison:** Compare short-, medium- and long-horizon estimates.

M07.02 Value at Risk (VaR)

Estimates a loss threshold that should not be exceeded at a selected confidence level over a specified horizon, subject to model assumptions.

- **Historical VaR:** Use the empirical distribution of portfolio returns or P&L.
- **Parametric VaR:** Estimate VaR from distributional assumptions and portfolio volatility/covariance.
- **Monte Carlo VaR:** Simulate portfolio outcomes from configured stochastic assumptions.
- **Model parameters:** Record confidence level, horizon, valuation base and lookback period for every result.

M07.03 Expected Shortfall

Estimates the average loss in the tail beyond the VaR threshold.

- **Conditional VaR:** Calculate average loss conditional on outcomes being beyond the VaR quantile.
- **Tail-loss estimation:** Measure severity of adverse outcomes rather than only the threshold.
- **Historical and simulated methods:** Compute tail averages from historical or Monte Carlo loss distributions.
- **Tail sample diagnostics:** Report the number and stability of observations used in the tail estimate.

M07.04 Drawdown Analytics

Measures losses from prior peaks and the time required to recover.

- **Maximum drawdown:** Find the largest peak-to-trough decline in the selected series.
- **Current drawdown:** Measure distance from the latest value to its prior running peak.
- **Recovery duration:** Measure time from a drawdown trough until the prior peak is regained.
- **Drawdown episodes:** Identify and summarize separate peak, trough and recovery periods.

M07.05 Correlation Analysis

Measures how asset or portfolio returns move together over a selected sample.

- **Asset correlation:** Calculate pairwise return correlations across holdings.
- **Rolling correlation:** Track changes in pairwise relationships over time.
- **Correlation matrix:** Expose a complete matrix for analysis and visualization.
- **Correlation regime flags:** Highlight unusually high or rapidly changing correlations.

M07.06 Covariance Analysis

Estimates joint return variability and supplies a core input to portfolio risk calculations.

- **Covariance matrix:** Calculate pairwise covariances using aligned return observations.
- **Portfolio covariance:** Combine asset covariances with portfolio weights.
- **Estimation window:** Configure lookback and observation frequency.
- **Matrix diagnostics:** Check symmetry, missing values and positive-semidefinite properties where required.

M07.07 Portfolio Beta

Measures sensitivity of a portfolio or security to a defined market benchmark.

- **Individual beta:** Estimate security sensitivity to benchmark returns.
- **Portfolio beta:** Aggregate or estimate the portfolio-level benchmark sensitivity.
- **Benchmark selection:** Use an explicitly selected and currency-compatible benchmark.
- **Regression diagnostics:** Report sample size, fit quality and estimation window.

M07.08 Sharpe & Sortino Ratios

Summarizes return relative to total or downside variability.

- **Sharpe ratio:** Compute excess return per unit of total return volatility.
- **Sortino ratio:** Compute excess return relative to downside deviation.
- **Risk-free rate:** Use a configured rate series consistent with return currency and frequency.
- **Period consistency:** Align return, risk-free and volatility frequencies before annualizing.

M07.09 Risk Contribution

Attributes total portfolio risk to individual holdings or groups.

- **Marginal risk contribution:** Estimate the change in portfolio risk from a small increase in a position weight.
- **Component risk contribution:** Allocate total risk across holdings using weights and marginal contributions.
- **Group contribution:** Aggregate contributions by sector, factor or portfolio sleeve.
- **Contribution reconciliation:** Check that component contributions reconcile to total risk within numerical tolerance.

M07.10 Risk Forecasting

Produces forward-looking estimates of risk using historical behavior and configurable statistical models.

- **Volatility forecasting:** Estimate future volatility using rolling or time-series models.
- **Rolling risk estimates:** Update risk metrics as new observations arrive.
- **Model comparison:** Compare alternative forecasts against realized outcomes.
- **Forecast evaluation:** Track error, bias and calibration over time; do not treat forecasts as guarantees.

M08 | Factor & Systematic Risk Analytics

Explains portfolio risk through common return drivers and separates broad systematic movements from security-specific behavior. Factor definitions and data sources must be versioned because factor models can materially change results.

Inputs: Security returns, factor-return series, benchmark returns, portfolio weights, classifications and model definitions.

Outputs: Factor exposures, factor sensitivities, systematic/idiosyncratic risk estimates and factor-level attribution.

8 top-level features

M08.01 Factor Definition

Defines the factor universe and the methodology used to construct or source each factor.

- **Market factor:** Represent broad equity-market return exposure.
- **Size factor:** Represent return differences associated with company size.
- **Value factor:** Represent return differences associated with valuation characteristics.
- **Momentum and quality:** Represent trend persistence and selected quality characteristics, with transparent construction rules.

M08.02 Factor Exposure

Measures how sensitive portfolio holdings are to selected factor returns.

- **Instrument loadings:** Estimate factor coefficients at security level.
- **Portfolio loadings:** Aggregate security exposures using portfolio weights.
- **Exposure normalization:** Standardize units and ensure factors are aligned across dates.
- **Exposure history:** Track changes in loadings and portfolio factor tilts.

M08.03 Factor Sensitivity

Uses statistical models to estimate how security or portfolio returns respond to factor movements.

- **Regression estimation:** Fit configured linear or other supported factor models.
- **Window configuration:** Select sample periods and observation frequency.
- **Coefficient diagnostics:** Report standard errors, significance and fit measures where applicable.
- **Stability checks:** Detect coefficient instability and insufficient sample coverage.

M08.04 Factor Attribution

Explains portfolio return or risk in terms of factor exposures and factor movements.

- **Factor return contribution:** Estimate return attributable to each factor over a period.
- **Risk attribution:** Estimate the risk contribution of factor exposures and factor covariance.
- **Residual contribution:** Report unexplained or idiosyncratic component separately.
- **Attribution reconciliation:** Show residuals when components do not fully reconcile because of model or data limitations.

M08.05 Systematic Risk

Estimates the portion of portfolio variability associated with common factors.

- **Common-factor variance:** Calculate variance explained by modeled factor movements.
- **Systematic component:** Aggregate factor-driven risk at instrument and portfolio level.
- **Scenario sensitivity:** Estimate factor impact under specified factor shocks.
- **Model coverage:** Report which holdings are covered by the factor model.

M08.06 Idiosyncratic Risk

Measures security-specific return variability not explained by the selected systematic factors.

- **Residual volatility:** Estimate volatility of model residuals.
- **Security-specific contribution:** Aggregate residual risk by holding.
- **Diversification view:** Show how issuer-specific risk contributes to total portfolio risk.
- **Residual diagnostics:** Flag outliers, nonstationarity or model misspecification indicators.

M08.07 Factor Correlation

Measures relationships among factor returns that influence combined factor risk.

- **Inter-factor matrix:** Calculate pairwise correlations among selected factors.
- **Rolling relationships:** Track correlations through time.
- **Factor covariance:** Supply covariance estimates to factor risk calculations.
- **Regime alerts:** Flag unusually high factor co-movement or breaks in historical relationships.

M08.08 Factor Monitoring

Continuously tracks factor exposures and changes that may alter the portfolio risk profile.

- **Exposure drift:** Compare current factor loadings with prior or target values.
- **Concentration flags:** Detect dependence on one or a small number of factors.
- **Threshold monitoring:** Evaluate factor exposure against configurable boundaries.
- **Trend reporting:** Show factor risk and exposure history.

M09 | Liquidity & Market Microstructure Risk

Evaluates whether positions can be reduced or exited at reasonable cost under current and stressed market conditions. The module uses order-book observations where available and clearly distinguishes displayed liquidity from executable or guaranteed liquidity.

Inputs: Order-book snapshots and updates, trades, quotes, position sizes, market sessions and execution observations.

Outputs: Spread, depth, imbalance, liquidity scores, market-impact estimates, slippage estimates and liquidation-cost/time measures.

9 top-level features

M09.01 Bid-Ask Spread

Measures the cost represented by the difference between best displayed bid and ask.

- **Absolute spread:** Calculate ask minus bid in price units.
- **Relative spread:** Scale the spread by midpoint or another documented reference.
- **Spread history:** Track intraday and historical spread behavior.
- **Spread quality checks:** Exclude stale or invalid books from spread calculations.

M09.02 Order Book Depth

Measures displayed quantities available at multiple price levels.

- **Bid/ask depth:** Aggregate displayed quantity on each side of the book.
- **Depth by level:** Report quantity and notional at each price level.
- **Depth imbalance:** Compare bid-side and ask-side displayed liquidity.
- **Depth persistence:** Measure how stable displayed depth is over a selected interval.

M09.03 Liquidity Estimation

Builds a multi-dimensional view of how easily an instrument may be traded.

- **Traded volume:** Measure executed quantity over selected intervals.
- **Turnover:** Measure value traded over time.
- **Liquidity scores:** Combine spread, depth, turnover and data availability into a documented score.
- **Liquidity buckets:** Classify instruments by estimated liquidity for screening and reporting.

M09.04 Market Impact

Estimates how an order of a given size could move the execution price.

- **Order-size sensitivity:** Evaluate estimated impact across proposed order sizes.
- **Depth-walk estimate:** Estimate execution cost by consuming displayed book levels.
- **Empirical impact models:** Use historical executions and market conditions to calibrate impact estimates.
- **Model assumptions:** Expose participation, horizon and market-regime assumptions.

M09.05 Slippage Estimation

Measures or estimates the difference between an intended reference price and an achieved execution price.

- **Expected price:** Define arrival price, midpoint or decision price as the benchmark.
- **Actual execution price:** Calculate volume-weighted execution price from fills.
- **Slippage attribution:** Separate spread, market movement, fees and residual execution effects where data permits.
- **Historical calibration:** Compare estimates with realized execution outcomes.

M09.06 Liquidity Stress

Tests how quickly liquidity can deteriorate under adverse conditions.

- **Spread widening:** Apply configurable spread multipliers or empirically observed shocks.
- **Depth reduction:** Reduce displayed or modeled depth under stress.
- **Volume contraction:** Model lower traded volume and participation capacity.
- **Stressed cost output:** Recalculate liquidation time and estimated cost under the stressed assumptions.

M09.07 Position Liquidation

Estimates the time and cost required to close a position under explicit execution assumptions.

- **Liquidation horizon:** Estimate sessions or intervals required at a configured participation rate.
- **Participation rate:** Limit modeled order flow to a share of market volume.
- **Liquidation cost:** Estimate spread, impact, fees and other configured cost components.
- **Partial liquidation:** Show cost and time for multiple reduction targets rather than only full exit.

M09.08 Order Book Imbalance

Measures relative buying and selling pressure in displayed book quantities.

- **Bid/ask pressure:** Compare bid and ask quantities over selected depth levels.
- **Imbalance indicators:** Calculate normalized imbalance metrics with documented conventions.
- **Time-series tracking:** Observe whether imbalance persists or reverses.
- **Signal limitations:** Treat imbalance as a microstructure descriptor, not a guaranteed directional prediction.

M09.09 Microstructure Monitoring

Monitors short-horizon market quality and abnormal changes in book behavior.

- **Spread volatility:** Measure variation in spreads across time.
- **Depth fluctuations:** Track rapid additions, cancellations and depletion of displayed depth.
- **Update intensity:** Measure event rates and periods of unusually high message activity.
- **Anomaly flags:** Detect abnormal book conditions, stale updates and sudden liquidity withdrawals.

M10 | Stress Testing & Scenario Analysis

Evaluates hypothetical and historical adverse conditions and measures their effect on portfolio value, exposure and liquidity. Each scenario must identify the shocks applied, valuation assumptions and limitations.

Inputs: Current or historical portfolio snapshots, market returns, factor data, correlation assumptions and scenario definitions.

Outputs: Stressed portfolio values, P&L and risk changes, scenario comparisons and loss attribution.

9 top-level features

M10.01 Historical Scenarios

Replays market conditions from selected historical periods against a current or historical portfolio.

- **Event library:** Maintain named historical episodes and their date ranges.
- **Historical shock mapping:** Apply observed asset or factor returns to current holdings using defined mapping rules.
- **Portfolio replay:** Revalue selected holdings under the event shocks.
- **Historical caveats:** Show differences between historical holdings and the portfolio being tested.

M10.02 Hypothetical Scenarios

Allows an analyst to define custom shocks for prices, factors, volatility or liquidity.

- **Price shocks:** Apply absolute or percentage shocks to selected securities or groups.
- **Volatility shocks:** Change volatility assumptions for risk calculations.
- **Custom correlations:** Modify correlation assumptions for scenario calculations.
- **Scenario templates:** Save reusable scenario definitions with version and author metadata.

M10.03 Market Crash Simulation

Models broad equity-market declines and evaluates portfolio sensitivity to severe market losses.

- **Index shocks:** Apply selected benchmark or broad-market shocks.
- **Beta-based response:** Estimate instrument changes using beta or other configured sensitivity models.
- **Nonlinear response:** Allow user-defined or calibrated stress adjustments beyond simple beta scaling.
- **Loss summary:** Report portfolio and position-level stressed P&L.

M10.04 Sector Shock Simulation

Evaluates concentrated adverse events affecting a sector or industry group.

- **Sector selection:** Select one or more sectors or industries to shock.
- **Sector-specific shocks:** Apply different shocks to selected groups.
- **Spillover assumptions:** Optionally apply linked shocks to related sectors or factors.
- **Contribution analysis:** Identify securities and groups driving the stressed result.

M10.05 Correlation Shock

Tests the effect of changing cross-asset relationships, particularly when diversification may weaken.

- **Correlation increase:** Move selected correlations toward a configured stress level.
- **Correlation convergence:** Test scenarios where correlations move toward one under stress.
- **Matrix validation:** Check that the shocked matrix is mathematically usable by the selected risk model.
- **Risk comparison:** Compare stressed and baseline portfolio risk.

M10.06 Liquidity Stress Testing

Combines market shocks with deteriorating trading conditions.

- **Spread stress:** Widen bid-ask spreads under the scenario.
- **Depth stress:** Reduce visible or modeled book depth.

- **Volume stress:** Reduce market volume or allowable participation.
- **Combined liquidation analysis:** Recalculate time and cost to exit positions under simultaneous market and liquidity shocks.

M10.07 Monte Carlo Scenarios

Simulates a distribution of possible portfolio outcomes based on configured stochastic models.

- **Random path generation:** Generate return paths using documented distributions and dependence assumptions.
- **Parameter configuration:** Set horizon, number of paths, volatility and correlation inputs.
- **Outcome distribution:** Summarize simulated P&L, quantiles and tail losses.
- **Reproducible seeds:** Store random seeds and model versions to make runs repeatable.

M10.08 Scenario Comparison

Allows multiple scenarios to be compared against one another and a baseline portfolio.

- **Baseline comparison:** Show changes relative to current portfolio valuation.
- **Scenario ranking:** Rank scenarios by loss, drawdown or another selected measure.
- **Sensitivity comparison:** Compare which assumptions drive differences between scenarios.
- **Saved results:** Persist run parameters and outputs for later review.

M10.09 Stress Loss Attribution

Explains where losses originate under each stress scenario.

- **Position contribution:** Attribute stressed P&L to individual holdings.
- **Sector contribution:** Aggregate losses by sector and industry.
- **Factor contribution:** Attribute modeled effects to factor shocks where supported.
- **Residual reporting:** Show unexplained effects and interaction terms when exact additive attribution is not possible.

M11 | Risk Limits & Control Framework

Defines risk boundaries and evaluates portfolio state or proposed orders against them. Breach severity, acknowledgement and remediation must be explicit; automated controls should fail safely when inputs are stale or unavailable.

Inputs: Risk metrics, exposures, liquidity estimates, limits, scenario results, portfolio state and proposed orders.

Outputs: Limit-utilization values, pre-trade decisions, breach events, escalation state and control audit records.

9 top-level features

M11.01 Portfolio Risk Limits

Defines portfolio-wide thresholds for key risk measures.

- **VaR limits:** Set absolute or NAV-relative VaR thresholds by confidence and horizon.
- **Volatility limits:** Set maximum permitted realized or forecast volatility.
- **Drawdown limits:** Define warning and hard limits for peak-to-trough portfolio decline.
- **Limit profiles:** Maintain different approved sets by portfolio or mandate.

M11.02 Position Limits

Restricts exposure to a single security or instrument.

- **Maximum position weight:** Set maximum permitted percentage of NAV.
- **Absolute notional limit:** Set currency-denominated exposure caps.
- **Long/short-specific limits:** Apply different thresholds by position direction.
- **Issuer aggregation:** Optionally aggregate multiple share classes or listings to issuer level.

M11.03 Sector Limits

Controls concentration in sectors or industries.

- **Sector caps:** Define maximum exposure by sector.
- **Industry caps:** Apply more granular limits.
- **Benchmark-relative limits:** Constrain active overweight or underweight where a benchmark is configured.
- **Classification-aware evaluation:** Handle unmapped or changing classifications explicitly.

M11.04 Liquidity Limits

Ensures that portfolio positions remain within defined liquidity tolerances.

- **Minimum liquidity:** Require a minimum liquidity score, depth or traded volume.
- **Maximum liquidation cost:** Set a cap on estimated cost to exit a position.
- **Maximum liquidation horizon:** Limit estimated time to liquidate under a configured participation rate.
- **Stressed liquidity thresholds:** Evaluate limits under selected stress assumptions.

M11.05 Dynamic Limits

Allows risk boundaries to vary with approved market or portfolio conditions.

- **Condition-based rules:** Change thresholds using documented volatility, liquidity or market-regime conditions.
- **Time-based profiles:** Apply different limits across sessions or scheduled periods.
- **Override governance:** Require reason, author and expiry for manual overrides.
- **Effective limit snapshot:** Record the exact threshold applied to each evaluation.

M11.06 Breach Detection

Evaluates risk state against active limits and identifies threshold violations.

- **Threshold comparison:** Compare current metrics with configured warning and hard boundaries.
- **Breach persistence:** Distinguish transient observations from sustained breaches where policy allows.

- **Stale-input checks:** Suppress or mark decisions when required data is stale or invalid.
- **Breach event creation:** Create a timestamped event with metric, threshold and portfolio context.

M11.07 Breach Classification

Classifies the severity and urgency of limit violations.

- **Warning:** Indicate proximity to a limit or a soft-threshold breach.
- **Critical:** Represent a material breach requiring prompt attention.
- **Emergency:** Represent a severe condition that may require immediate operational controls.
- **Classification policy:** Keep severity mapping configurable and versioned.

M11.08 Risk Escalation

Manages the workflow after a breach is raised.

- **Acknowledgement:** Record that an authorized user has reviewed the event.
- **Escalation:** Route unresolved or worsening breaches to a higher responsibility level.
- **Resolution tracking:** Record action plan, owner, due time and closure reason.
- **Breach lifecycle:** Maintain open, acknowledged, mitigated, closed and reopened states.

M11.09 Pre-trade Risk Checks

Evaluates proposed orders before they are passed to an execution system.

- **Order validation:** Check instrument, side, quantity, price and required metadata.
- **Post-trade exposure simulation:** Estimate portfolio state after the proposed order or fill.
- **Limit evaluation:** Recalculate relevant position, sector, gross/net, VaR and liquidity constraints.
- **Decision response:** Return pass, reject or review-required with reason codes; never treat missing risk inputs as an automatic pass.

M12 | Risk Alerts & Notification Engine

Turns important changes and control events into actionable alerts. Alert generation should be deduplicated, prioritized and linked to the underlying metric or breach so an analyst can investigate quickly.

Inputs: Risk metrics, thresholds, feed-health events, liquidity indicators, portfolio changes and user notification preferences.

Outputs: Alert events, notification messages, acknowledgement and resolution state, searchable alert history.

8 top-level features

M12.01 Threshold Alerts

Creates alerts when configured risk measures approach or cross thresholds.

- **VaR alerts:** Notify on warning or hard VaR threshold changes.
- **Volatility alerts:** Notify when realized or forecast volatility exceeds configured levels.
- **Exposure alerts:** Notify on instrument, sector, gross or net exposure breaches.
- **Drawdown alerts:** Notify when portfolio drawdown reaches warning or critical levels.

M12.02 Market Alerts

Detects unusual market behavior that may change portfolio risk.

- **Abnormal price movements:** Flag large changes relative to configured history or market bands.
- **Volatility spikes:** Detect abrupt increases in short-horizon volatility.
- **Trading halts:** Surface instrument halts or suspension events when the feed provides them.
- **Market-wide events:** Aggregate broad market alerts for instruments held by the portfolio.

M12.03 Liquidity Alerts

Raises alerts when market quality deteriorates for held securities.

- **Spread widening:** Detect widening relative to baseline or configured thresholds.
- **Depth deterioration:** Detect sudden or persistent decreases in displayed depth.
- **Liquidity score deterioration:** Notify when composite liquidity measures fall below tolerance.
- **Liquidation risk:** Alert when estimated exit cost or horizon breaches a configured limit.

M12.04 Portfolio Alerts

Highlights portfolio-specific events that warrant review.

- **Large losses:** Alert on material intraday or period P&L declines.
- **Concentration increases:** Notify when holdings or sectors become more concentrated.
- **Position changes:** Surface unusual position or cash movements.
- **Reconciliation breaks:** Notify when external and internal books diverge materially.

M12.05 Alert Prioritization

Ranks alerts so users can focus on the most important and time-sensitive conditions.

- **Severity:** Use warning, critical and emergency classifications.
- **Urgency:** Consider how quickly the condition may require a response.
- **Impact:** Estimate affected portfolio value, risk or number of instruments.
- **Deduplication and grouping:** Combine repeated alerts for the same underlying condition without hiding escalation.

M12.06 Alert Management

Provides a workflow for reviewing and resolving alerts.

- **Acknowledgement:** Record reviewer and acknowledgement time.
- **Assignment:** Assign alerts to an owner or role.

- **Comments and disposition:** Capture investigation notes and selected actions.
- **Closure:** Close resolved alerts with a reason and preserve the complete event history.

M12.07 Notification Channels

Delivers selected alerts through configured channels.

- **Dashboard notifications:** Show in-app alert counts and details.
- **Email integration:** Send selected alert classes to configured recipients.
- **Delivery controls:** Apply quiet periods, retry policies and channel-specific severity rules.
- **Delivery status:** Record sent, failed or retried state without losing the original alert.

M12.08 Alert History

Stores historical alert data for audit and analysis.

- **Search and filters:** Search by portfolio, instrument, severity, type and time range.
- **Event timeline:** Show trigger, acknowledgement, escalation and closure times.
- **Recurring alert analysis:** Identify repeated conditions and noisy rules.
- **Export:** Provide controlled export for operational review.

M13 | Dashboard & Visualization Layer

Presents portfolio state and risk analysis in a form that supports rapid monitoring and investigation. The interface is a consumer of backend APIs; it must not contain authoritative risk calculations or control decisions.

Inputs: Portfolio and risk APIs, market-data status, alert services, scenario outputs and user layout preferences.

Outputs: Interactive dashboards, charts, tables, heatmaps, drill-down views and user-selected filters.

10 top-level features

M13.01 Portfolio Overview

Provides a concise view of portfolio size, performance and directional exposure.

- **Total NAV:** Display net asset value with valuation timestamp and base currency.
- **P&L:** Show intraday and period profit/loss with clear calculation basis.
- **Gross and net exposure:** Display absolute and directional market exposure.
- **Data freshness:** Expose when the underlying portfolio and market values were last updated.

M13.02 Risk Overview

Surfaces the principal risk measures used for daily portfolio monitoring.

- **VaR:** Show selected confidence/horizon VaR and model method.
- **Expected Shortfall:** Show average modeled loss in the tail.
- **Volatility:** Display realized or forecast volatility with lookback window.
- **Drawdown:** Display current and maximum drawdown with selected period.

M13.03 Exposure Dashboard

Allows users to inspect where capital and risk are concentrated.

- **Sector allocation:** Show sector weights and market values.
- **Instrument allocation:** Show ranked holdings and short exposures.
- **Portfolio allocation:** Show allocation across positions and cash.
- **Drill-down filters:** Filter from portfolio to sector, industry and individual security.

M13.04 Risk Heatmaps

Uses visual encoding to identify clusters of risk or concentration.

- **Sector heatmap:** Compare sector-level exposure, P&L or risk.
- **Security heatmap:** Compare holdings by selected risk measure.
- **Color scale controls:** Make metric direction and scale explicit.
- **Drill-through:** Open the underlying holdings and values behind a cell.

M13.05 Correlation Dashboard

Visualizes return relationships among selected holdings or factors.

- **Correlation matrix:** Render pairwise correlations for a selected universe.
- **Time-window selection:** Change lookback and sampling frequency.
- **Cluster visibility:** Group securities with similar correlation patterns where supported.
- **Outlier inspection:** Inspect unexpectedly high or changing relationships.

M13.06 Liquidity Dashboard

Displays current and historical trading conditions for portfolio holdings.

- **Bid-ask spreads:** Show absolute and relative spread.
- **Order-book depth:** Display depth by side and price level.
- **Liquidity scores:** Show composite liquidity and component measures.

- **Liquidation estimates:** Show modeled cost and time to reduce positions.

M13.07 Stress Testing Dashboard

Presents scenario definitions and their estimated effect on the portfolio.

- **Scenario results:** Show stressed value and P&L by scenario.
- **Loss distribution:** Visualize simulated outcome quantiles where Monte Carlo is used.
- **Scenario comparison:** Compare alternative shocks and assumptions.
- **Loss attribution:** Identify leading holdings, sectors or factors behind stressed losses.

M13.08 Alerts Dashboard

Gives users a consolidated view of current risk events.

- **Active breaches:** Show unresolved limit breaches by severity.
- **Recent alerts:** Display newly triggered market, liquidity and portfolio events.
- **Alert actions:** Allow authorized users to acknowledge, assign or close alerts.
- **Alert filters:** Filter by portfolio, severity, type and time.

M13.09 Historical Analytics

Supports analysis of how portfolio and risk measures evolved.

- **Historical portfolio values:** Chart NAV and P&L over time.
- **Risk trends:** Chart VaR, volatility, drawdown and exposure history.
- **Event overlays:** Annotate material alerts, trades or scenario events.
- **Point-in-time selection:** Load historical states with their associated data and model versions.

M13.10 Custom Views

Allows users to tailor layouts without changing the underlying calculation logic.

- **Saved layouts:** Save widget arrangement and selected metrics.
- **Filters and time ranges:** Persist common portfolio, instrument and date filters.
- **Widget configuration:** Choose display fields and supported visualization settings.
- **Sharing controls:** Define whether a view is personal or available to an authorized group.

M14 | Reporting, Attribution & Audit

Produces durable reports and evidence for portfolio review, control oversight and historical investigation. Reports must carry the portfolio timestamp, selected methodology and data/model versions used to generate them.

Inputs: Portfolio snapshots, risk outputs, scenarios, alerts, audit events and report configuration.

Outputs: Daily and ad hoc reports, exports, attribution summaries, audit trails and archived report versions.

8 top-level features

M14.01 Daily Risk Reports

Produces a repeatable daily view of portfolio risk and control status.

- **Portfolio summary:** Include NAV, P&L and gross/net exposure.
- **Risk metrics:** Include VaR, Expected Shortfall, volatility and drawdown.
- **Limit status:** Summarize active warnings and breaches.
- **As-of metadata:** Include report date, valuation time, currency and data freshness.

M14.02 Portfolio Performance Reports

Explains portfolio return and P&L over selected periods.

- **Return series:** Report period and cumulative returns using defined cash-flow treatment.
- **P&L breakdown:** Separate realized and unrealized components where available.
- **Benchmark comparison:** Compare against a selected benchmark with aligned dates and currency.
- **Performance caveats:** Disclose missing data, external cash flows and valuation limitations.

M14.03 Risk Attribution Reports

Explains which holdings, groups or factors contribute to measured risk.

- **Asset contribution:** Show security-level risk contribution.
- **Sector contribution:** Aggregate risk contribution by sector and industry.
- **Factor contribution:** Show systematic factor contribution where the model supports it.
- **Reconciliation:** Explain rounding, residuals or non-additive effects.

M14.04 Stress Testing Reports

Documents scenario assumptions and stressed portfolio outcomes.

- **Scenario description:** State shock magnitudes, affected groups and horizon.
- **Stressed P&L:** Show total and position-level estimated loss.
- **Sensitivity summary:** Compare outcomes under alternative assumptions.
- **Limit implications:** Highlight any stressed limits exceeded by the result.

M14.05 Liquidity Reports

Summarizes current and stressed ability to exit portfolio positions.

- **Liquidity profile:** Report spread, depth, volume and liquidity scores.
- **Liquidation estimates:** Show expected time and cost under stated participation assumptions.
- **Concentration of illiquidity:** Identify positions contributing most to portfolio liquidation risk.
- **Stress comparison:** Show how estimates change under liquidity shocks.

M14.06 Export Engine

Exports authorized report and analytical data in common formats.

- **CSV:** Export flat data for programmatic analysis.
- **Excel:** Export structured worksheets for analyst review.
- **PDF:** Generate paginated reports with methodology and as-of metadata.

- **Export controls:** Apply permissions, field selection and output validation.

M14.07 Audit Trail

Records material user and system actions to support traceability.

- **User actions:** Record configuration edits, approvals and operational commands.
- **System events:** Record job executions, feed transitions and control decisions.
- **Configuration changes:** Link changes to user, timestamp, reason and version.
- **Tamper resistance:** Restrict modification and deletion of audit records through application permissions and retention controls.

M14.08 Historical Report Archive

Preserves generated reports so prior decisions can be reviewed consistently.

- **Report versioning:** Store each generated report as a distinct version.
- **Retrieval:** Search by portfolio, report type, period and run ID.
- **Run metadata:** Store model, data and configuration versions with the report.
- **Retention:** Apply documented storage and retention policies.

M15 | API, Security & Infrastructure

Supplies the shared runtime, interfaces, access controls and operational safeguards required by all modules. This is a cross-cutting platform capability and should be built incrementally rather than deferred until the end.

Inputs: Application services, user identities, environment configuration, secrets, database settings and operational policies.

Outputs: REST and streaming interfaces, authenticated sessions, persistent storage, logs, metrics, backups and deployable local services.

12 top-level features

M15.01 Backend API

Exposes stable application services to the dashboard and authorized integrations.

- **REST endpoints:** Provide versioned resource-oriented endpoints for portfolios, risk, scenarios and controls.
- **Request validation:** Validate types, ranges, identifiers and business invariants at the boundary.
- **Response schemas:** Return consistent response structures, pagination and error codes.
- **API documentation:** Publish machine-readable OpenAPI documentation and examples.

M15.02 Real-time API

Streams selected updates to connected clients and services.

- **WebSocket channels:** Expose authenticated channels for market, portfolio, risk and alert updates.
- **Subscription management:** Allow clients to subscribe to permitted portfolios and instruments.
- **Message sequencing:** Include event IDs or sequence metadata to support client resynchronization.
- **Connection lifecycle:** Handle heartbeat, reconnect, authorization expiry and bounded client queues.

M15.03 Database Infrastructure

Persists transactional, historical and time-series information using fit-for-purpose storage patterns.

- **PostgreSQL:** Store portfolios, positions, configuration, limits, alerts and application metadata.
- **Time-series storage:** Store high-volume observations using partitioning or a dedicated time-series extension where justified.
- **Analytical storage:** Separate large analytical workloads from latency-sensitive transactional operations when necessary.
- **Data lifecycle:** Define indexes, retention, archival, migrations and backup strategy.

M15.04 Authentication

Verifies user identity before granting access to the application.

- **Login:** Support a secure local authentication flow or trusted identity provider.
- **Session management:** Issue, expire and revoke sessions safely.
- **Password handling:** Use a modern password-hashing function and never store plaintext passwords.
- **Authentication events:** Log successful, failed and revoked authentication events.

M15.05 Authorization

Determines which users can view or operate on each resource.

- **Role-based access control:** Define roles such as viewer, analyst, operator and administrator.
- **Resource permissions:** Restrict access by portfolio, module and action.
- **Least privilege:** Grant only the minimum permissions required.
- **Authorization testing:** Verify that denied actions cannot be performed through alternate API paths.

M15.06 Encryption

Protects sensitive information in storage and during communication.

- **Secrets management:** Store API keys and database credentials outside source code.

- **Transport security:** Use TLS where services communicate across a network boundary.
- **Sensitive-field protection:** Encrypt or avoid persisting highly sensitive configuration values.
- **Key rotation:** Support rotation and revocation of credentials and encryption keys.

M15.07 System Monitoring

Measures the operational health of the application and its dependencies.

- **CPU and memory:** Track resource utilization and saturation.
- **Database monitoring:** Track connection pools, query latency, locks and storage growth.
- **Process monitoring:** Track service uptime, worker status and restart counts.
- **Service health checks:** Expose readiness and liveness signals for local operations.

M15.08 Logging

Creates structured records that explain application behavior and failures.

- **Structured logs:** Use consistent fields such as timestamp, service, severity and correlation ID.
- **Error logs:** Capture exceptions with relevant context while excluding secrets.
- **Request tracing:** Correlate API requests with downstream processing and jobs.
- **Log retention:** Rotate, retain and protect logs according to operational needs.

M15.09 Backup & Recovery

Protects application data and provides tested recovery procedures.

- **Database backups:** Create scheduled backups with documented scope.
- **Recovery procedures:** Restore data and services in a controlled order.
- **Recovery testing:** Periodically test restore integrity rather than assuming backups work.
- **Recovery objectives:** Define acceptable recovery time and data-loss windows for the local deployment.

M15.10 Testing Infrastructure

Provides repeatable checks for correctness, reliability and performance.

- **Unit tests:** Test calculations, validators and business rules in isolation.
- **Integration tests:** Test database, API, feed and module interactions.
- **Performance tests:** Measure throughput, latency and resource behavior under representative loads.
- **Regression tests:** Use fixed datasets and expected outputs to detect unintended changes.

M15.11 Deployment

Packages and starts the local application consistently across development environments.

- **Local environment:** Define reproducible runtime and dependency configuration.
- **Configuration management:** Separate environment-specific settings from application code.
- **Service startup:** Coordinate API, workers, database and frontend startup dependencies.
- **Release process:** Version builds and apply database migrations in a controlled manner.

M15.12 Operational Safety

Ensures failures or unsafe states do not silently propagate into risk decisions or execution pathways.

- **Kill switch:** Provide an authorized way to disable selected live-capable integrations or order flow.
- **Fail-safe behavior:** Block or mark calculations and controls when required inputs are invalid or unavailable.
- **Execution isolation:** Keep pre-trade checks and execution adapters separated from ordinary dashboard presentation.
- **Operational runbooks:** Document shutdown, recovery, degraded-mode and incident procedures.

Module dependencies and information flow

The dependencies below describe the normal data flow. They do not prevent shared infrastructure from being developed in parallel. M15 provides cross-cutting services to all modules, even though it is shown at the end of the list.

M01 -> M02: Feed Registry supplies validated connection and subscription configuration to the ingestion engine.

M02 -> M03: Ingestion delivers raw and normalized market observations for quality checks.

M03 + M04 -> M05: Validated market data and canonical instrument/reference data support portfolio valuation and position processing.

M05 -> M06 -> M07: Positions and exposure weights feed the core portfolio risk calculations.

M02 -> M09: Order-book and trade events feed spread, depth, imbalance and liquidity analysis.

M07 -> M08 and M10: Core risk outputs and returns support factor analysis and scenario/stress calculations.

M07 + M09 + M10 -> M11: Risk, liquidity and scenario outputs are evaluated against limits and controls.

M11 -> M12 -> M13: Breach events become alerts and are surfaced in the dashboard.

M05-M13 -> M14: Portfolio, risk, stress, alert and audit information feeds reporting and archival.

M15 -> all modules: APIs, storage, security, monitoring, logging, testing and deployment support every module.

Recommended implementation phases

Phase 1 - Foundation and data: M01-M04 plus essential M15 capabilities. Deliver feed configuration, ingestion, validation, instrument master, database migrations, logging and health monitoring.

Phase 2 - Portfolio core: M05-M06. Deliver position ledger, valuation, reconciliation, portfolio snapshots and exposure analytics.

Phase 3 - Risk intelligence: M07-M09. Deliver core statistical risk, factor analytics and order-book liquidity measures with test datasets.

Phase 4 - Risk controls: M10-M12. Deliver stress scenarios, limits, pre-trade risk evaluation, breach workflows and alerts.

Phase 5 - Dashboard: M13. Deliver portfolio overview, risk, exposure, liquidity, scenario and alert views backed by APIs.

Phase 6 - Reporting and hardening: M14 plus remaining M15. Deliver report generation, audit archive, performance tests, backup/restore tests and operational runbooks.

Suggested MVP boundary

A practical first release should include feed configuration, one reliable market-data source, data-quality checks, instrument master, portfolio and position management, exposure analytics, a baseline risk engine and a basic dashboard. It should also include authentication, structured logging, database backups and enough monitoring to diagnose failures. Advanced factor models, Monte-carlo scenarios, full order-book history and automated live order routing can follow after the core data and portfolio ledger are validated.

Acceptance criteria applicable across modules

Correctness: Calculations and transformations have unit tests against hand-checked or independent reference cases.

Traceability: Outputs include timestamps, identifiers, source references and relevant model/configuration versions.

Failure handling: Invalid inputs and unavailable dependencies produce explicit errors or degraded states rather than silent success.

Idempotency: Retries and replays do not duplicate trades, market events, portfolio changes or alert events.

Security: Secrets are excluded from source code and logs; API permissions are tested at the resource/action level.

Performance: Latency and throughput are measured against representative order-book and portfolio workloads.

Operability: Each service exposes health, logs and recovery steps; backup restoration is tested.

Auditability: Material configuration changes, control decisions and user actions can be reconstructed.

Important assumptions and boundaries

Market-data licensing: Real-time exchange or vendor feeds may require paid entitlements. The architecture does not imply that public APIs provide complete or exchange-authorized depth data.

Order-book scope: Order-book analytics require the appropriate depth feed and accurate sequence handling. A REST quote API or delayed snapshot is not equivalent to a full event-level order-book feed.

Risk-model limitations: VaR, Expected Shortfall, factor models, beta and liquidity estimates are model outputs, not guarantees. Distributional assumptions, lookback windows and data gaps must be disclosed.

Execution boundary: Pre-trade validation can be developed and tested independently. Live execution requires separate broker/exchange adapters, account permissions, kill-switch testing, monitoring and operational approvals.

Accounting conventions: Cost basis, cash treatment, corporate actions, FX conversion and benchmark alignment must be selected explicitly before performance and P&L are treated as authoritative.

Regulatory posture: This specification is an engineering design document, not legal, regulatory, investment or compliance certification.